

A counterexample to the simultaneous-rotation orbit lemma in arXiv:1307.3206

Candidate full counterexample packet

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Status: candidate counterexample, likely valid pending expert review.

Abstract

Dann and Zymonopoulou assert that the orbit of every vector in $\mathbb{R}^{\kappa n}$ under simultaneous block rotation by $\text{SO}(\kappa)$ is a $(\kappa - 1)$ -sphere whose linear span has dimension κ . We give a minimal counterexample: for $\kappa = 3$, $n = 2$, the orbit of (e_1, e_2) spans all of $\mathbb{R}^3 \oplus \mathbb{R}^3$. More generally, if the n blocks form a $\kappa \times n$ matrix X of rank r , then for every $\kappa \geq 3$ the orbit span has dimension κr . Consequently the paper's subspace $H_x^\perp$ has variable dimension and the ensuing fixed-codimension “hyperplane” and intersection-body construction does not work as stated. The separately formulated L_{-p} embedding question remains open; we do not claim a resolution of it.

1 Source assertion

The source is S. Dann and M. Zymonopoulou, *Intersection Bodies with Certain Symmetries*, arXiv:1307.3206v2, published under the title *Sections of Convex Bodies with Symmetries* [1]. Write a vector $x \in \mathbb{R}^{\kappa n}$ as n consecutive blocks $v_1, \dots, v_n \in \mathbb{R}^\kappa$. The paper defines

$$R_\sigma(x) = (\sigma v_1, \dots, \sigma v_n), \quad \sigma \in \text{SO}(\kappa).$$

Lemma 1 on PDF page 4 asserts that $\{R_\sigma(x) : \sigma \in \text{SO}(\kappa)\}$ is always a $(\kappa - 1)$ -sphere, that its span $H_x^\perp$ always has dimension κ , and that $H_x = (H_x^\perp)^\perp$ always has dimension $\kappa n - \kappa$.

with $x \in \mathbb{R}^{\kappa n}$. For $x \in \mathbb{R}^{\kappa n}$, $\rho_K(x) = \|x\|_K$, is the Euclidean distance from the origin to the boundary of K in the direction x . The set of κ -balanced star bodies in $\mathbb{R}^{\kappa n}$ forms the class of *star bodies* in $\mathbb{R}^{\kappa n}$.

Now we introduce the notion of a hyperplane in $\mathbb{R}^{\kappa n}$. For a vector $y \in \mathbb{R}^{\kappa n}$, denote by $|y|_2$ the Euclidean length of y . Denote by e_i the vector in $\mathbb{R}^{\kappa n}$ with the i -th coordinate equal to one and all other coordinates equal to zero.

Lemma 1. Fix $\kappa \geq 2$ and a vector $x \in \mathbb{R}^{\kappa n}$. The set $\{R_\sigma(x) : \sigma \in \text{SO}(\kappa)\}$ is a $(\kappa - 1)$ -dimensional sphere of radius $|x|_2$ and its span, denoted by $H_x^\perp$, is a κ -dimensional subspace of $\mathbb{R}^{\kappa n}$. Moreover, the orthogonal complement of $H_x^\perp$, denoted by H_x , is a κ -balanced $(\kappa n - \kappa)$ -dimensional subspace of $\mathbb{R}^{\kappa n}$.

Proof. It is enough to establish the claim for any convenient vector x . Choose $x = (0, \dots, 0, 1)$, then $\{R_\sigma(x) : \sigma \in \text{SO}(\kappa)\}$ is the unit sphere of the κ -dimensional subspace spanned by the vectors $e_{\kappa(n-1)+1}, \dots, e_{\kappa n}$. $\square$

For an element $\xi \in S^{\kappa n-1}$, we call H_ξ the *hyperplane* in $\mathbb{R}^{\kappa n}$ determined by the vector ξ .

Figure 1: Lemma 1 and its proof in arXiv:1307.3206v2, PDF page 4.

The proof checks only the special rank-one vector $(0, \dots, 0, 1)$. Simultaneous left rotation is not transitive on the unit sphere of $\mathbb{R}^{\kappa n}$: it preserves the Gram matrix of the blocks, and in particular their linear rank.

2 Idea of the counterexample

Regard x as the matrix $X = [v_1 \cdots v_n] \in M_{\kappa, n}(\mathbb{R})$. Then $R_\sigma(x)$ is simply σX . When X has two independent columns, its orbit retains their mutual inner products and is an orbit of orthonormal frames, not a sphere. For $X = [e_1 \ e_2] \in M_{3,2}(\mathbb{R})$, six rotations give orbit points whose sums and differences are the six coordinate vectors of $\mathbb{R}^3 \oplus \mathbb{R}^3$.

The general calculation is equally short. For $\kappa \geq 3$, the linear span of the group $\text{SO}(\kappa)$ is the full matrix algebra $M_\kappa(\mathbb{R})$. Hence the orbit span is $\{AX : A \in M_\kappa(\mathbb{R})\}$. Each of its κ rows can be chosen independently in the row space of X , giving dimension $\kappa \text{rank } X$.

3 The concrete counterexample

Proposition 1. *Let $\kappa = 3$, $n = 2$, and $X = [e_1 \ e_2]$. Then*

$$\text{span}\{\sigma X : \sigma \in \text{SO}(3)\} = M_{3,2}(\mathbb{R}) \cong \mathbb{R}^6.$$

In particular the orbit is not a 2-sphere spanning a 3-plane.

Proof. For every ordered orthonormal pair (u, v) in $\mathbb{R}^3$, the matrix with columns $u, v, u \times v$ belongs to $\text{SO}(3)$. Thus the following six pairs are orbit points:

$$(e_1, e_2), (e_1, -e_2), (e_2, e_3), (e_2, -e_3), (e_3, e_1), (e_3, -e_1).$$

Taking half-sums and half-differences of the two points in each pair gives

$$(e_1, 0), (0, e_2), (e_2, 0), (0, e_3), (e_3, 0), (0, e_1).$$

These are the six coordinate vectors of $\mathbb{R}^3 \oplus \mathbb{R}^3$, so the orbit span is all of $\mathbb{R}^6$. A 2-sphere in a linear 3-plane cannot have a six-dimensional span. $\square$

Indeed, the orbit is the Stiefel manifold of ordered orthonormal 2-frames in $\mathbb{R}^3$, which is diffeomorphic to $\text{SO}(3)$ and has dimension 3, rather than a 2-sphere.

4 General corrected orbit-span formula

Theorem 2. *Let $\kappa \geq 3$, let $X \in M_{\kappa, n}(\mathbb{R})$ have rank r , and let $W \subseteq \mathbb{R}^n$ be its row space. Then*

$$\text{span}\{\sigma X : \sigma \in \text{SO}(\kappa)\} = \{Y \in M_{\kappa, n}(\mathbb{R}) : \text{every row of } Y \text{ belongs to } W\}.$$

Consequently

$$\dim \text{span}\{\sigma X : \sigma \in \text{SO}(\kappa)\} = \kappa r,$$

and the orthogonal complement has dimension $\kappa(n - r)$.

Proof. First we show that $\text{span SO}(\kappa) = M_\kappa(\mathbb{R})$. Let

$$\mathcal{D} = \{\text{diag}(\varepsilon_1, \dots, \varepsilon_\kappa) : \varepsilon_j \in \{-1, 1\}, \prod_j \varepsilon_j = 1\} \subset \text{SO}(\kappa).$$

For $\kappa \geq 3$, character cancellation in this finite sign group gives

$$E_{jj} = \frac{1}{|\mathcal{D}|} \sum_{D_\varepsilon \in \mathcal{D}} \varepsilon_j D_\varepsilon.$$

For each i, j , choose $Q_{ij} \in \text{SO}(\kappa)$ with $Q_{ij}e_j = e_i$. Since left multiplication by Q_{ij} preserves the linear span of the group,

$$E_{ij} = Q_{ij}E_{jj} \in \text{span SO}(\kappa).$$

All matrix units therefore belong to this span.

It follows that

$$\text{span}\{\sigma X : \sigma \in \text{SO}(\kappa)\} = \{AX : A \in M_\kappa(\mathbb{R})\}.$$

Every row of AX is a linear combination of the rows of X , hence belongs to W . Conversely, if every row of Y belongs to W , choose the coefficients expressing each row of Y in terms of the rows of X and assemble them as the rows of a matrix A ; then $Y = AX$. The displayed space consists of κ independently chosen rows in the r -dimensional space W , so its dimension is κr . Its orthogonal complement has dimension $\kappa n - \kappa r = \kappa(n - r)$. $\square$

Corollary 3. *For $\kappa \geq 3$, the dimension asserted in source Lemma 1 holds only when X has rank one (apart from the zero vector). For the rank-two matrix in Proposition 1, the source's notation gives $H_x^\perp = \mathbb{R}^6$ and $H_x = \{0\}$, not two complementary 3-planes.*

5 Consequences and scope

Immediately after Lemma 1, the paper calls H_ξ a hyperplane in $\mathbb{K}^n$ for every unit vector ξ and defines intersection bodies using the volumes of $D \cap H_\xi^\perp$ and $L \cap H_\xi$. The next paragraph requires $D \cap H_\xi^\perp$ to be a κ -dimensional ball. Theorem 2 shows instead that the two dimensions are κr and $\kappa(n - r)$, where r is the rank of the block matrix of ξ . They vary with ξ , and for $r > 1$ the first section need not be a κ -ball. Thus the defect is structural, not merely a faulty parametrization of a correct sphere.

Later parallel-section arguments that take $H = H_\xi$ with fixed codimension κ , or identify the orbit with the unit sphere of $H_\xi^\perp$, need a new definition or an additional rank-one restriction. In particular, the printed proof of the universal embedding range for $\kappa \geq 3$ cannot be accepted from this route without repair.

The source also asks whether every origin-symmetric κ -balanced convex body embeds in L_{-p} for $p \in (\kappa n - \kappa, \kappa n - 3)$ when $\kappa \geq 4$. That statement is well-defined independently of the subspaces H_ξ . This packet neither proves nor disproves it; the full result here is the counterexample to the foundational orbit lemma and the exact replacement for its span formula.

6 Novelty check

The run's cheap indexes contained no packet for arXiv:1307.3206 or this orbit calculation. Bounded searches on 9 August 2026 used the exact lemma, source title and identifier, and combinations with “correction”, “erratum”, “simultaneous rotation”, and “ κ -balanced”. They found the source and its published record but no correction. This supports possible novelty but is not exhaustive bibliographic certification.

References

- [1] S. Damm and M. Zymonopoulou, *Sections of Convex Bodies with Symmetries*, Adv. Math. 271 (2015), 112–152; arXiv:1307.3206v2.